Equations, Ratio & Similarity

ANSWER KEY

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Answers with the key steps. Problem numbers match the practice set.

1. Solving Equations

1. **EASY** **Answer:** $x = 14$

   Work: $x = 23 - 9 = 14$

2. **EASY** **Answer:** $x = 7$

   Work: $x = 42 \div 6 = 7$

3. **EASY** **Answer:** $x = 20$

   Work: $\frac{x}{4} = 5 \quad x = 20$

4. **MEDIUM** **Answer:** $x = -6$

   Work: $-2x = 12 \quad x = -6$

5. **HARD** **Answer:** $x = 18$

   Work: $\frac{2}{3}x = 12 \quad x = 12 \cdot \frac{3}{2} = 18$

2. Multi-Step Equations

6. **EASY** **Answer:** $x = 4$

   Work: $2x + 6 = 14 \quad 2x = 8,\ x = 4$

7. **EASY** **Answer:** $x = 5$

   Work: $5x - 4 = 21 \quad 5x = 25,\ x = 5$

8. **EASY** **Answer:** $x = 4$

   Work: $2x + 1 = 9 \quad 2x = 8,\ x = 4$

9. **MEDIUM** **Answer:** $x = 6$

   Work: $6x - 3 = 4x + 9 \quad 2x = 12,\ x = 6$

10. **HARD** **Answer:** $x = 11$

    Work: Multiply both sides by 4: $3x - 1 = 2(x + 5) \quad 3x - 1 = 2x + 10 \quad x = 11$

3. Literal Equations

11. **EASY** **Answer:** $t = \frac{d}{r}$

    Work: Divide both sides by r.

12. **EASY** **Answer:** $w = \frac{A}{l}$

    Work: Divide both sides by l.

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13. EASY **Answer:** $b = y - x$

    Work: Subtract x from both sides.

14. MEDIUM **Answer:** $x = \dfrac{y-b}{m}$

    Work: $y - b = mx \quad x = \dfrac{y-b}{m}$

15. HARD **Answer:** $b_2 = \dfrac{2A}{h} - b_1$

    Work: $2A = h(b_1 + b_2) \quad \dfrac{2A}{h} = b_1 + b_2 \quad b_2 = \dfrac{2A}{h} - b_1$

4. Ratio and Proportion

16. EASY **Answer:** $3 : 5$

    Work: $15 \div 5 = 3,\ 25 \div 5 = 5$

17. EASY **Answer:** $x = 3$

    Work: $20x = 60 \quad x = 3$

18. EASY **Answer:** $\$3$ **per notebook**

    Work: $18 \div 6 = 3$

19. MEDIUM **Answer:** 40 **girls**

    Work: $3k + 5k = 64,\ k = 8 \quad$ girls $= 5 \cdot 8 = 40$

20. HARD **Answer:** 24 **blue marbles**

    Work: red $= 3k,$ blue $= 4k \quad 3k + 6 = 4k \Rightarrow k = 6 \quad$ blue $= 4 \cdot 6 = 24$ (red: $18 + 6 = 24$)

5. Similar Figures

21. EASY **Answer:** 3

    Work: $15 \div 5 = 3$

22. EASY **Answer:** $40°$

    Work: Matching angles of similar figures are equal.

23. EASY **Answer:** 10

    Work: $\dfrac{6}{3} = \dfrac{x}{5} \quad x = 10$

24. MEDIUM **Answer:** 27 **feet**

    Work: $\dfrac{6}{4} = \dfrac{x}{18} \quad 4x = 108,\ x = 27$

25. HARD **Answer:** 45 **square units**

    Work: Areas scale by the square: $\left(\dfrac{3}{2}\right)^2 = \dfrac{9}{4} \quad 20 \cdot \dfrac{9}{4} = 45$

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